

STUDY INFORMATION SHEET

Study Information

IRB reference number: IRB-2018-09-002

Protocol Title: Acoustics & Speech in Singapore – Intuitions by Singaporeans

Title of study: Decisions about words

Student Investigator: Tan Kai Jie

Principal Investigator & Contact Details:

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Objective of the Research Study:

You are invited to participate in a research study. This investigational study is designed to find out what words sound like in Singapore. You will be asked to complete one or more of the following tasks: 1) Recording of audio, and 2) Decisions about words. You are invited because you are a healthy adult with typical hearing, with a specific set of language experiences. This study will recruit up to 20 participants per group, recruited online or on NTU Campus.

It is important to us that you first take time to read through and understand the information provided in this sheet. Before you take part in this research study, the study will be explained to you and you will be given the chance to ask questions. When you feel that you understand this study, and that you wish to take part in the study, you must sign the informed consent form. You will be given a copy of this information sheet to take home with you if you like.

Are there any exclusion criteria?

You are invited because you are a healthy adult with typical hearing, who has a particular language background

What procedures will be followed in this study?

What will I be asked to do?

If you decide to take part in this study, you will be asked to take part in tasks that allow us to find out more language use in your community. Your participation in the study will last around less than 10 minutes, in a single session. This study will involve decisions about words, as listed below.

Decisions about words

You will be asked to make simple decisions about different words or sounds (written or spoken), which may include word pairs, or groups of words. These decisions might be about whether two words sound the same or different, about how to spell different words, or whether two words rhyme. You will be asked to make some simple decisions about written or spoken words, or simple sounds. All of the words you encounter will be familiar words from daily life, made up words (called 'pseudowords'), or impossible words (called 'non-words'). None of the words or sounds will contain offensive or unpleasant content. You will be asked to make the decisions in the following tasks. The precise tasks will be described in more detail below – as indicated by ticks.

☐ Pronunciation intuitions task. You will be asked to share your intuitions about the pronunciation of written English words. For example, you might be asked whether two written words are pronounced the same or different (e.g., 'four' vs 'floor').

☐ Audio discrimination task. You will be asked to listen to pre-recorded audio of words (e.g., 'four'), non-words (e.g., 'bloor'), or other sounds, and to make simple decisions about whether different items sound the same or different. For example, people might be asked "Are these two words the same or different: 'four' vs 'floor'?" or "Out of these three words, does the middle one sound the same as the first or the second: 'sis' 'sess' 'sess'?"

☐ Audio transcription task. You will be asked to listen to pre-recorded audio recordings of human speech and write down what word you think you have heard. In some versions of the task, you may be asked to write just one word, or as many words as you think the sounds might represent (e.g., 'right' and 'write'). This task may be accompanied by a confidence rating (e.g., "How confident are you that this is what the speaker said?")

☐ Audio matching task. You will be asked to listen to audio recordings of human speech and indicate whether what you heard matches a

specific word, presented visually. For example, you may be asked, “Did you hear this word: ‘tree?’” or “Does this word match this picture? [picture of a tree]” Visual stimuli will be written words or images of physical objects (e.g. a tree). These visual stimuli will contain no offensive or unpleasant content.

☐ Typicality rating. You will be asked to listen to audio recordings of words and then rate the audio tokens for their typicality. For example, “On a scale from 1-7 rate how typical this speaker sounds when saying this word, use 0 if you think they are saying a different word”.

☐ Rhyme evaluation task. You will be asked to listen to audio recordings of words and then indicate whether the words rhyme (e.g. ‘speech’ ‘peach’- Do these words rhyme?).

☐ Rhyme production task. You will be asked to say/write a word that rhymes with a word (e.g. Write a word that rhymes with ‘tree’).

☐ Pun evaluation task. You will be asked to evaluate whether different words can be used to create puns according to the way that they sound (e.g., name of a bakery: ‘Bread Pitt’).

☐ Lexical decision task. You will be shown a sequence of letters on a screen or played a short recording of human speech and asked to decide if what you saw/heard is a real word or not.

☒ Local word inventory. For a given list of words, you will be asked a series of questions about how familiar you are with the words, what you think they mean, and how common you think the words are among young Singaporeans (for e.g., Singaporeans 3 years and younger).

Demographic information. At the end of the task you will be asked some questions that will help us characterize our participant group. You will be asked about your age, what country you grew up in, and some questions about what languages you speak. You will also be asked some questions about your sensory experiences (e.g., eyesight and hearing).

Right to Refuse or Withdraw. Your participation is completely voluntary. Before the study begins, you will be fully informed of what the study involves, and will be allowed to ask any questions about the study. You may decide not to participate, to discontinue participation, or skip any part of the task, at any time, without penalty. Please note that if you skip a task, it might not be possible to include your data in the complete analysis of the study, meaning that your contribution to science may not have as much impact. You are also free to withdraw from these studies at any time, without penalty.

Possible Risks and Discomforts. There are no anticipated risks associated with participating in this study, beyond those encountered in daily life (speaking into a microphone and using a computer, or pen and paper). The Internal Review Board (IRB) of

Nanyang Technological University has reviewed and approved the study.

Possible Benefits from Participating in the Study. There are no direct benefits for participating. However, participation in the research carried out by NTU researchers can provide you with the opportunity to acquire first-hand knowledge of the research process. In addition, some participants find that taking part in studies about language provide them with interesting insights into their day-to-day language use.

Costs & Payments if Participating in the Study.

There is no remuneration for taking part in this online quiz. You will bear no costs for this study, the cost of the tasks will be borne by the grants to PI Suzy Styles.

Confidentiality of Study.

All data collected will be anonymous. After participating in the study you will be asked if you want your audio to be tagged with a unique ID known to you.

How will my privacy be protected?

The researcher conducting the study has received training in ethical practices and procedures before beginning the research project. They have learned about the importance of informed consent, and ethical considerations surrounding privacy – especially as these concerns apply to those under the age of 21.

Data collected at the time of the study will include an anonymous numeric code, so that your responses in different parts of the study can be connected to each other. The only non-anonymous record of your participation will be the consent document. The numerical code may be used to link your name to your responses in the study, by use of a ‘key’. The consent form and the record of sensory experiences will be stored securely in a locked cabinet administered by the Project PI (Asst Prof Suzy Styles). Medical data will be used primarily for demographic purposes – e.g.

to describe the characteristics of the population who took part. The raw, anonymous data will include links between an individual’s sensations and their answers in the survey, but no uniquely identifying information like name, or contact information. This raw data may be shared with the PIs collaborators both inside and outside Singapore.

So that your participation can make the widest possible contribution to science, at the end of the study, the raw data will be archived in a data repository that will be accessible to scientists all over the world. Before we do this, this key linking data to the names of individuals who took part in the study will be destroyed, making the data completely anonymous. After the key has been destroyed it will no longer be possible to request your data be removed from the study.

For the purpose of oversight, the NTU Institutional Review Board will be allowed direct access to the original study materials, including personal information you have provided during this study. This is so the IRB can check study procedures and data. They will do this without making any of your information public.

Personal Data.

Your participation in this study will involve the collection of data in an individually-identifiable form (or “Personal Data”).

“Personal Data” means data about you which makes you identifiable: (i) from such data; or (ii) from that data and other information which an organisation has or likely to have access. This includes medical conditions, medications, investigations and treatment history.

Data collected are the property of NTU. In the event of any publication regarding this study, your identity will remain confidential. Information collected for this study will be kept confidential and stored for a minimum of 10 years in a secure environment with restricted access within NTU. Your records, to the extent of the applicable laws and regulations, will not be made publicly available.

Only anonymized research data will be transferred out of Singapore to the PIs collaborators during the study, and for later data repository.

By signing the Informed Consent Form attached, you are authorizing (i) collection, access to, use and storage of your “Personal Data”, and (ii) disclosure to, and use and storage by, authorised service providers and relevant third parties, whether located in Singapore or overseas, for the purposes of the study. Research arising in the future, based on this “Personal Data”, will be subject to review by the relevant institutional review board.

Additional information:

What will happen to the results of the study?

This information will be scientifically analysed and prepared for scientific dissemination, which may include presentation in international journals, and at academic conferences. The researchers will also make the research findings available on the BLIP website after formal peer review has taken place.

The results may also be used for illustration of the research in a scientific context, for example, presentation at international

scientific conferences, use in university lectures, or public science demonstrations by the PI. Similarly, if the research results in peer-reviewed journal publication, they may be published in a journal article, or hosted on the research web-pages. In all cases, no identifying information about you will be included: individuals will only be described in terms of age (in years), along with general background information.

Open Science Promise: You are offering to donate your time to science. To honour your contribution, we plan to make the results available, so that other scientists can learn from them. During the project, this may include sharing data with a broad team of collaborators at different universities in different countries (as described above). At the end of the project, we plan to make each decision made by each person available for scientists all over the world. We will only share anonymous, de-identified information, so no-one will be able to know who was involved in the study, or who made which decision, but they will be able to find out information like “the person who answered ‘A’ for question 1 speaks three languages”. By sharing all of the information, your contribution can have the greatest possible impact on our scientific understanding of sensory processes. We plan to make the anonymous, de-identified data available in a stable, long-term repository such as the Open Science Framework of the Center for Open Science, or DR-NTU (Data) – Nanyang Technological University’s in-house digital data repository.

Possible future contact: In the event of any new information becoming available (including but not limited to occurrences of serious adverse events, and changes in proposed research) that may be relevant to your willingness to continue in this study, you (or your legally acceptable representative, if relevant) will be informed in a timely manner by the Principal Investigator (or his/her representative) and will be contacted for further consent if required. Any individually-identifiable data obtained during the course of this study will be stored and used only for the purposes of this study. Your Personal Data will not be used for future research, unless otherwise consented by you in the accompanying Consent Form.

Who to contact if You have Questions:

The study has been reviewed by NTU Institutional Review Board (the central ethics committee) for ethics approval. For questions, complaints or feedback about the research, you are encouraged to contact the Principle Investigator Asst Prof Suzy Styles: suzy.styles@ntu.edu.sg. You may also contact the Institutional Review Board at NTU, who approved this study.

If you want an independent opinion to discuss problems and questions (complaints / feedback), obtain information and offer inputs on your rights as a research subject, please contact:

NTU-Institutional Review Board
Research Integrity and Ethics Office
62 Nanyang Drive
N1.2-B1-02A
Singapore 637459
Email: irb@ntu.edu.sg

INFORMED CONSENT FORM

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What is the study about? The purpose of this study is to investigate Decisions about words. If you take part, you will be asked to make simple decisions about written or spoken words. For example, you may be asked to indicate whether you think a pair of words sound the same or different. The tasks in the study will include computer based tasks or pencil and paper tasks. You will also be asked some questions about your language background, to help us understand your responses. You might be asked to imagine different communicative scenarios before giving your responses.

This is a plain language summary of what the study will involve.
More detailed information can be found on the attached pages.

Consent To Take Part in the Study

I have read, discussed and understand the information and procedures in the study information sheet attached to this consent form. My questions concerning the study have been answered to my satisfaction, and I acknowledge that I am participating in this study of my own free will. I understand that I may refuse to participate or stop participating at any time

Name of Participant

Signature

Date

Contact for Future Research

Your identity and contact details will not will not be used to contact you for future research